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College of Engineering – Department of Civil Engineering

CURRICULUM

Bachelor of Science in Geodetic Engineering

Academic Year 2025-2026

Reference CMOs: CHED Memorandum Order No. 89 S. 2017, CMO No. 39, series of 2021, No. 27, series of 2015, No. 20, series of 2013

Curriculum Description

The Bachelor of Science in Geodetic Engineering is designed to provide students the knowledge and background of collecting and measuring spatial data above, on, or below the surface of the earth using appropriate technologies and the scientific and methodological processing and management of these data for tor the production of spatial information

The program also aims to develop graduates with ethical behavior, integrity, social responsibility, effective communication, teamwork, and leadership skills, while promoting lifelong learning to keep up with the continuing technological advancement.

Graduates of this program are expected to be engaged in conduct of various surveys including the establishment of geodetic control network; collection of ground data using various methodologies, techniques, platforms and sensors; processing, evaluation and analysis of collected data to generate information for various applications; quality assurance of outputs in accordance with the accepted standards and specifications; development of survey and mapping standards and protocols; conduct of research and development activities; development of spatial information systems; and development of business entrepreneurial skills.

Program Education Objectives

The geodetic engineering alumni three to five years after graduation shall:
PEO 1: **Specialist**
PEO 2: **Professionalism and Leadership**
PEO 3: **Lifelong Learning**

Outcomes

The following skills, knowledge, and values are expected to be attained by students as they progress through the program:

SO1 Discipline Knowledge. Ability to apply mathematics, sciences and principles of engineering to solve complex geodetic engineering problems;
SO2 Investigation. Ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions;
SO3 Design/Development of Solutions. Design solution, system, components, processes, exhibiting improvements/innovations, that meet specified needs with appropriate consideration for public health and safety, cultural, societal, economical, ethical, environmental and sustainability issues.
SO4 Leadership and Teamwork. Function effectively as a member of a leader on a diverse team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.
SO5 Problem Analysis. Identify, formulate, and solve complex geodetic engineering problems by applying principles of engineering, science, and mathematics;

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SO6 Ethics and Professionalism. Apply ethical principles and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, environmental, and societal contexts.

SO7 Communication. Communicate effectively on complex geodetic engineering activities with the community, and the society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions;

SO8 Environment and Sustainability. Recognize the impact of professional engineering solutions in societal, global, and environmental contexts and demonstrate knowledge of and need for sustainable development;

SO9 Lifelong Learning. Recognize the need for, and ability to engage in independent and life-long learning in the broadest context of technological change.

SO10 The Engineer and Society. Apply reasoning based on contextual knowledge to assess societal, health, safety, legal, cultural, contemporary issues, and the consequent responsibilities relevant to professional engineering practices.

SO11 Modern Tool Usage. Apply appropriate techniques, skills, and modern engineering and IT tools to complex geodetic engineering activities;

SO12 Project Management and Finance. Demonstrate knowledge and understanding of engineering management and financial principles as member or a leader of a team to manage projects in multidisciplinary settings and identify opportunities of entrepreneurship.

SO13 Social and National Responsibility. Apply acquired engineering knowledge and skills in addressing community problems that contributes to national development.

CURRICULUM COMPONENTS

Classification / Field / Course	Number of Hours per Week		Credit Units
	Lec	Lab	
I. TECHNICAL COURSES			
A. Mathematics			
Differential Calculus	4	0	4
Integral Calculus	4	0	4
Differential Equations	3	0	3
Probability and Statistics for AI	2	3	3
Subtotal	**13**	**3**	**14**
B. Natural/ Physical Sciences			
Chemistry for Engineers	3	3	4
Physics for Engineers	3	3	4
Modern Biology	2	3	3
Subtotal	**7**	**9**	**11**
C. Basic Engineering Sciences			
Introduction to Engineering	0	3	1
Engineering Drawing	0	3	1
Computer Programming	1	6	3
Computer-Aided Design	1	3	2
Engineering Economics	3	0	3
Engineering Management	3	0	3
Basic Occupational Safety and Health	3	0	3
Technopreneurship	3	0	3
Engineering Mechanics	3	0	3
Subtotal	**17**	**15**	**22**
D. Allied Courses			

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Geology	3	0	3
Electrical and Electronics Engineering for Geodetic Engineers	2	3	3
Advanced Information & Communications Technology	3	0	3
Environmental Science and Engineering	3	0	3
Subtotal	**17**	**3**	**12**
E. Professional Courses			
General Surveying 1	2	3	3
General Surveying 2	2	6	4
Cartography	2	3	3
Engineering Surveys	2	6	4
Geometric Geodesy	3	0	3
Introduction to the Laws on Public and Private Lands	2	0	2
Theory of Errors and Adjustments	3	0	3
Geodetic Engineering Laws, Obligations and Contracts, Ethics	2	0	2
Physical Geodesy	3	0	3
Satellite Geodesy	3	0	3
Geodetic Surveying	2	6	4
Geodetic Computations & Adjustments	2	6	4
Remote Sensing	2	6	4
Land Registration Laws	3	0	3
Fundamentals of Geographic Information Systems	1	6	3
Land Use Planning and Development	2	3	3
Property Surveys	3	6	5
Hydrographic Surveying	2	3	3
Photogrammetry	2	6	4
Survey Camp	0	3	1
Public Land Laws & Laws on Natural Resources	3	0	3
Special Studies in Geodetic Engineering	1	6	3
Methods of Research	3	0	3
Land Administration, Management and Valuation Systems	3	0	3
Geodetic Engineering Orientation	1	0	1
Capstone Design Project 1	0	3	1
Capstone Design Project 2	0	3	1
GE Practice with Comprehensive Examination	0	6	2
Subtotal	**54**	**81**	**81**
F. Elective Courses			
Geovisualization	3	0	3
Subtotal	**3**	**0**	**3**
F. On-the-Job Training			
OJT/Internship (320 hrs)	320		4
Subtotal	**320**		**4**
TOTAL TECHNICAL COURSES	**111**	**111**	**147**
II. NON-TECHNICAL COURSES			
A. General Education Courses			
Understanding the Self	3	0	3
Mathematics in the Modern World	3	0	3
The Contemporary World	3	0	3

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Readings in Philippine History	3	0	3
Purposive Communication	3	0	3
Ethics	3	0	3
Art Appreciation	3	0	3
Science, Technology and Society	3	0	3
Subtotal	**24**	**0**	**24**
B. Mandated and General Education Elective Courses			
Life and Works of Rizal	3	0	3
Linear Algebra for AI	3	0	3
Machine Learning and Neural Networks	2	3	3
The Entrepreneurial Mind	3	0	3
Subtotal	**11**	**3**	**12**
C. Physical Activities Toward Health and Fitness			
Movement Competency Training	2	0	2
Exercise-based Fitness Activities	2	0	2
Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor, and Adventure Activities 1	2	0	2
Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor, and Adventure Activities 2	2	0	2
Subtotal	**8**	**0**	**8**
D. National Service Training Program			
National Service Training Program 1	3	0	3
National Service Training Program 2	3	0	3
Subtotal	**6**	**0**	**6**
TOTAL NON-TECHNICAL COURSES	**148**	**3**	**50**
GRAND TOTAL	**259**	**114**	**197**

SUMMARY

Courses	Number of Units
I. TECHNICAL COURSES	
A. Mathematics	14
B. Natural/Physical Sciences	11
C. Basic Engineering Sciences	22
D. Allied Courses	12
E. Professional Courses	81
F. Geodetic Engineering Elective Course	3
G. On-the Job training	4
II. NON-TECHNICAL COURSES	
A. A. Required General Education	24
B. B. Mandated and Elective Courses	12
C. C. Physical Activities Toward Health and Fitness (PATHFit)	8
D. D. National Service Training Program	6
GRAND TOTAL	**197**

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CURRICULUM STRUCTURE

FIRST YEAR						
FIRST SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**
GEd 102	Mathematics in the Modern World	3	3	0		
GEd 105	Readings in Philippine History	3	3	0		
GEd 101	Understanding the Self	3	3	0		
CHEM 101	Chemistry for Engineers	4	3	3		
GEd 106	Purposive Communication	3	3	0		
ENGG 101	Introduction to Engineering	1	0	3		
MATH 101	Differential Calculus	4	4	0		
PATHFit 1	Movement Competency Training	2	2	0		
NSTP 111	National Service Training Program 1	3	3	0		
	Total	**26**	**24**	**6**		

FIRST YEAR						
SECOND SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**
MATH 102	Integral Calculus	4	4	0	MATH 101	
PHYS 101	Physics for Engineers	4	3	3	MATH 101	
GEd 104	The Contemporary World	3	3	0		
GEd 109	Science, Technology and Society	3	3	0		
GEd 108	Art Appreciation	3	3	0		
CpE 101	Computer Programming	3	1	6		
ENGG 102	Engineering Drawing	1	3	0		
PATHFit 2	Exercise-based Fitness Activities	2	2	0	PATHFit 1	
NSTP 121	National Service Training Program 2	3	3	0	NSTP 111	
	Total	**26**	**25**	**9**		

FIRST YEAR						
MIDTERM						
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**
BioE 101	Modern Biology	3	2	3	PHYS 101	
GEd 103	Life and Works of Rizal	3	3	0		
GEd 107	Ethics	3	3	0		
	Total	**9**	**8**	**3**		

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SECOND YEAR						
FIRST SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**
MATH 103	Differential Equations	3	3	0	MATH 102	
GE 201	Geodetic Engineering Orientation	1	1	0		
PATHFit 3	Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor, and Adventure Activities 1	2	2	0	PATHFit 1 & 2	
ENGG 103	Computer-Aided Design	2	1	3		
ENGG 109	Engineering Mechanics	3	3	0	PHYS 101	
GE 202	General Surveying 1	3	2	3		
SCI 405	Geology	3	3	0	PHYS 101	
GE 203	Cartography	3	2	3		
AI 101	Linear Algebra for AI	3	3	0	MATH 102	
	Total	**23**	**20**	**9**		

SECOND YEAR						
SECOND SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**
GE 204	General Surveying 2	4	2	6	GE 202	
GE 206	Engineering Surveys	4	2	6		GE 204
GE 207	Geometric Geodesy	3	3	0		GE 204
GE 208	Electrical and Electronics Engineering for Geodetic Engineers	3	2	3	PHYS 101	
GE 209	Introduction to the Laws on Public and Private Lands	2	2	0		
GE 210	Theory of Errors and Adjustments	3	3	0	MATH 103	
PATHFit 4	Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor, and Adventure Activities 1	2	2	0	PATHFit 1 & 2	
AI 102	Probability and Statistics for AI	3	2	3	AI 101	
GEdE 101	The Entrepreneurial Mind	3	3	0		
	Total	**27**	**21**	**18**		

THIRD YEAR						
FIRST SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**
GE 301	Geodetic Engineering Laws, Obligations and Contracts, Ethics	2	2	0	GE 203	
ENGG 116	Methods of Research	3	3	0	MATH 103	
GE 302	Physical Geodesy	3	3	0	GE 207	

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GE 303	Satellite Geodesy	3	3	0	GE 207	
GE 304	Geodetic Surveying	4	2	6	GE 207	GE 302
GE 305	Geodetic Computations & Adjustments	4	2	6	GE 210	GE 302
GE 306	Remote Sensing	4	2	6	PHYS 101	
AI 103	Machine Learning and Neural Networks	3	2	3		
	Total	**26**	**19**	**21**		

THIRD YEAR						
SECOND SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**
GE 307	Land Registration Laws	3	3	0	GE 209	
GE 308	Fundamentals of Geographic Information Systems	3	1	6	GE 203	
GE 309	Land Use Planning and Development	3	2	3		GE 308
GE 310	Property Surveys	5	3	6	GE 202	
GE 311	Hydrographic Surveying	3	2	3	GE 202	
GE 312	Photogrammetry	4	2	6	GE 305, GE 203	
ENGG 104	Engineering Economics	3	3	0	MATH 102	
ENGG 106	Engineering Management	3	3	0	4th yr Standing	
	Total	**27**	**19**	**24**		

THIRD YEAR						
MIDTERM						
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**
ENGG 114	Basic Occupational Safety and Health	3	3	0		
ENGG 113	Environmental Science and Engineering	3	3	0		
GE 313	Survey Camp	1	0	3	GE 204	
	Total	**7**	**6**	**3**		

FOURTH YEAR						
FIRST SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**
ENGG 117	OJT/Internship (320 hrs)	4	320 hrs		4th yr standing	
ENGG 420	Capstone Design Project 1	1	3	0	ENGG 116	
	Total	**5**	**3**	**0**		

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FOURTH YEAR						
SECOND SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**
ENGG 421	Capstone Design Project 2	1	3	0	ENGG 420	
ENGG 105	Technopreneurship	3	3	0	4th yr Standing	
GE 401	Public Land Laws & Laws on Natural Resources	3	3	0	GE 309	
GE 402	Geovisualization	3	3	0		GE 401
GE 403	Advanced Information & Communications Technology	3	3	0	4th yr Standing	
GE 404	Land Administration, Management and Valuation Systems	3	3	0	4th yr Standing	
GE 405	Special Studies in Geodetic Engineering	3	1	6	4th yr Standing	
GE 406	GE Practice with Comprehensive Examination	2	0	6	Graduating Standing	
	Total	**21**	**19**	**12**		